Dominic Bair

dbair@tennessee.edu   www.dbair.xyz
https://github.com/DothWalrus   https://linkedin.com/in/dominic-bair-861ab0189/

Education

- 2022 – · · · **Ph.D. candidate, Mathematics,** University of Tennessee Knoxville (UTK).
- 2020 – 2022 **M.S. Mathematics,** Montana State University (MSU).
- 2016 – 2020 **B.S. Mathematics; Minor: Physics,** Montana State University.

Publications

1. D. Bair, S. Jana, and Y. Qing, "First-passage percolation, non-positive curvature, and radial maps," *Preprint*, 2025. URL: https://arxiv.org/abs/2512.05720.
2. D. Bair, "DVM: A deep learning algorithm for minimizing functionals," *Theses and Dissertations at Montana State University (MSU)*, 2022. URL: https://scholarworks.montana.edu/xmlui/handle/1/16840.
3. C. Potts, J. Schearer, T. A. Sebrell, *et al.*, "MNPmApp: An image analysis tool to quantify mononuclear phagocyte distribution in mucosal tissues," *Cytometry Part A*, vol. 101, no. 12, pp. 1012–1026, 2022. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/cyto.a.24657.

Employment History

Paid Positions

- 2022 – . . . **Graduate Teaching Assistant,** UTK Department of Mathematics.
- 2023 – 2024 **Climbing Wall Attendant/Setter,** UTK RECSports.
- 2020 – 2022 **Graduate Teaching Assistant,** MSU Department of Mathematical Sciences.
- 2015 – 2022 **Laborer,** 3 Brothers Plumbing and Heating.
- 2020 **Laborer,** Peak Property Management.
- 2019 – 2020 **Undergraduate Research Assistant,** MSU Department of Mathematical Sciences.
- 2018 – 2019 **Undergraduate Research Assistant,** MSU Department of Physics.

Volunteer Positions

- 2026 – . . . **Senator,** UTK Graduate Student Senate.
- **Leader,** UTK Math Graduate Student Seminar.
- 2018 – 2020 **Assistant,** National Center for Unwanted Firearms.
- 2015 – 2017 **Firefighter,** Clancy Volunteer Fire Department.

Courses Taught

- M115 **Statistical Reasoning,** UTK.
- M125 **Basic Calculus,** UTK.
- M113 **Mathematical Reasoning,** UTK.
- M119 **College Algebra,** UTK.
- M171 **Calculus I,** MSU.
- M151 **Precalculus,** MSU.
- M161 **Survey of Calculus,** MSU.

Employment History (continued)

M121 — **College Algebra,** MSU.

Research Interests

My research interests lie at the intersection of geometric group theory, probability, dynamical systems, and applied stochastic modeling. I am currently studying how finite-state automata and Markov measures describe geodesics and boundary measures in relatively hyperbolic groups, with particular interest in Patterson–Sullivan theory. I am also interested in random metric geometry, including how first-passage percolation changes hyperbolicity, Morse geodesics, and other manifestations of nonpositive curvature. Alongside these theoretical questions, I use stochastic calculus, numerical simulation, and statistical calibration to investigate derivative pricing and hedging, including quanto options. Broadly, I seek structural principles connecting geometry, randomness, dynamics, and computation across pure and applied mathematics.

Skills

Coding/Software — Codex, MATLAB, Python, LaTeX, Julia, Windows, MacOS, Linux.

Misc. — Technical problem solving, leadership, academic research, teaching, first aid.

Miscellaneous Experience

Awards and Achievements

2026
- **Elected as UTK Graduate Student Senator**.
- **UTK Graduate Student Academic Achievement Award**.

2025
- **UTK Oral Specialty Exam,** passed.
- **UTK Preliminary Exam in Topology,** passed.
- **UTK Preliminary Exam in Stochastics,** passed.

2020
- **NSF GRFP Reviews,** Excellent, Excellent, Very Good. (Fellowship not received).
- **NSF and AWM Travel Grant,** to present at Data Science and Image Analysis Conference of the Pacific Northwest.

2019 – 2020
- **INBRE Grant,** to support the Spatial Statistics in Histology Images research. NIH funded, MSU based grant

2016 – 2020
- **MSU Premier Scholarship,** for demonstration of academic promise, leadership experience and other honors.

2019
- **Superior Achievement in the Field of Mathematics,** recognition by the National Honorary Mathematics Society, $\Pi M E$, for superior achievement in the field of mathematics.

2016
- **Eagle Scout,** the highest achievement that can be obtained in the Scouts of America.

Talks Given

2026
- **Ordered frequency in relatively hyperbolic groups,** UTK Topology Seminar.
- **First-passage percolation, non-positive curvature, and radial maps,** YGGT XIV.
- **Graphons and random graphs,** UTK Graduate Student Seminar.

2025
- **Introduction to first-passage percolation and subadditive processes,** UTK Probability Seminar.
- **First-passage percolation on hyperbolic graphs: bi-infinite geodesics, radial maps, and infinitely many thick triangles,** UTK Topology Seminar.

2021
- **Deep variational methods,** MSU Poster Slam.
- **Deep variational methods,** Walk the Walk Chalk the Chalk.

Miscellaneous Experience (continued)

2020 **Spatial statistics in histology images,** SIAM Northern States Section Student Chapters Conference.

Spatial statistics in histology images, Data Science and Image Analysis Conference of the Pacific Northwest.

2019 **Spatial statistics in histology images,** MSU Winter Research Celebration.

Reading Courses

2026 **Big Mapping Class Groups,** UTK.

Fundamentals of Stein's Method (Ross), UTK.

2023 **Monte Carlo Strategies in Scientific Computing (Liu),** UTK.

Certifications

2026 **Quant Finance Bootcamp,** The Erdös Institute.

2023 **CPR and First Aid,** American Red Cross.